

EOGS++: Earth Observation Gaussian Splatting with Internal Camera Refinement and Direct Panchromatic Rendering

Supplementary Material

gardiens.github.io/EOGS2

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1. Additional visual results

We presents additional visual results in Figure 1.

2. Elevation reconstruction accuracy without the foliage mask

We provide the complete elevation reconstruction accuracy without the foliage mask in Table 1.

References

Billouard, C., Derksen, D., Sarrazin, E., Vallet, B., 2024. Satingp : Unleashing neural graphics primitives for fast relightable transient-free 3d reconstruction from satellite imagery.

Marí, R., Facciolo, G., Ehret, T., 2023. Multi-date earth observation nerf: The detail is in the shadows. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2035–2045.

Savant Aira, L., Facciolo, G., Ehret, T., 2025. Gaussian Splatting for Efficient Satellite Image Photogrammetry. *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 5959–5969.

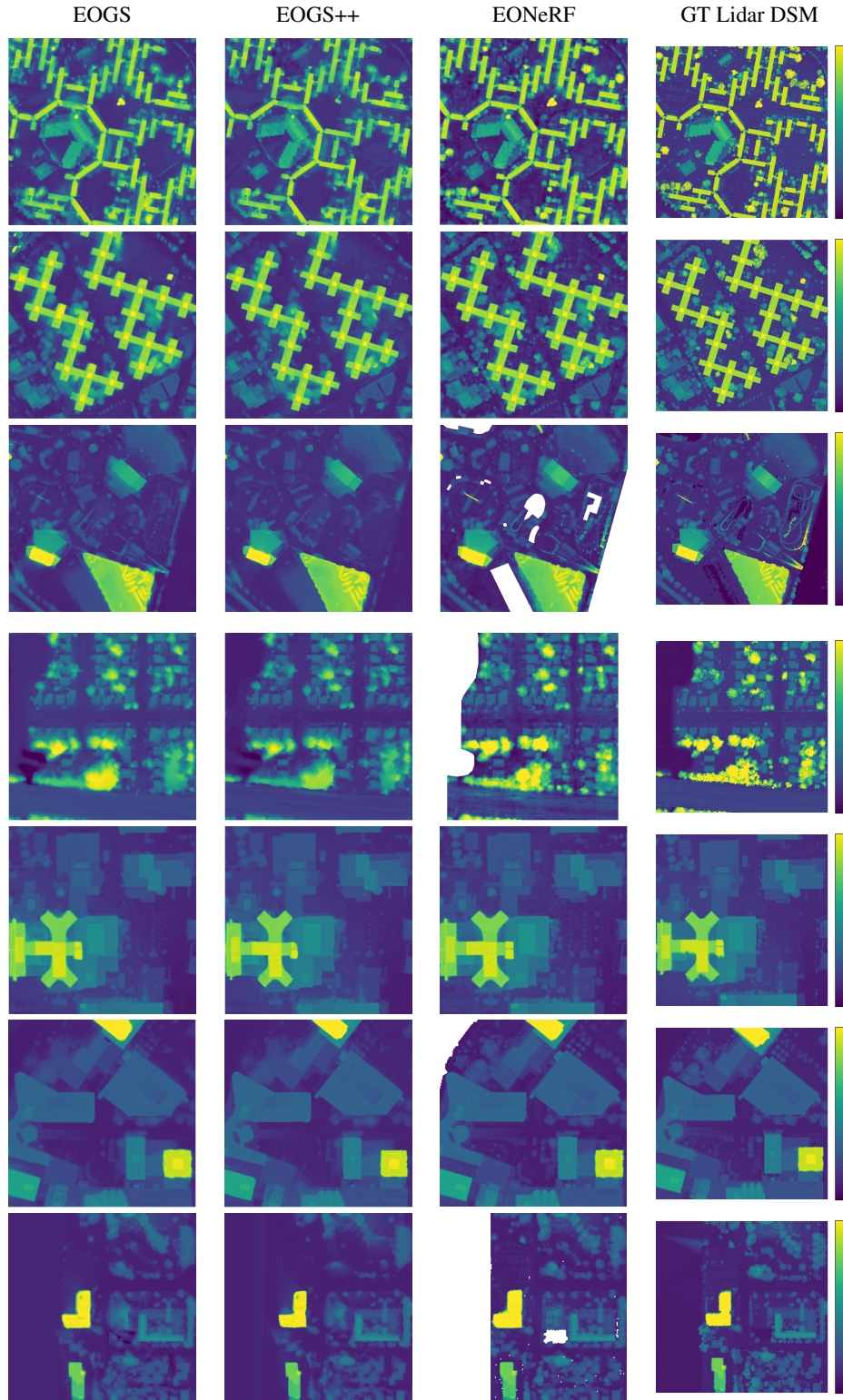


Figure 1. Qualitative comparison of EOGS, EOGS++, and EONeRF predicted DSMs against the GT LiDAR DSM on the respective IARPA 001 IARPA 002 IARPA 003 JAX 004 JAX068 JAX 214 and JAX 260 scenes.

	B.A. RPC Brovey Image	Opacity Reset TSDF	JAX					IARPA				Pooled mean ↓	Time [min] ↓
			004	068	214	260	mean ↓	001	002	003	mean ↓		
EOGS	✓	✗	1.46	1.09	1.72	1.47	1.43	1.56	2.0	1.25	1.61	1.525	9
SAT-NGP	✓	✗	1.63	1.27	2.18	1.79	1.72	1.54	2.11	1.69	1.78	1.75	25
EO-NeRF	✓	✗	1.37	1.05	1.61	1.37	1.35	1.43	1.79	1.31	1.51	1.43	900
EOGS++	✓	✓	1.54	0.99	1.55	1.52	1.40	1.55	1.86	1.42	1.61	1.52	25
EOGS++	✗	✓	1.84	0.99	1.81	1.93	1.64	1.78	2.02	1.66	1.82	1.73	32

Table 1. Quantitative comparison of EOGS++ and baseline methods on elevation reconstruction accuracy and runtime **without the foliage mask**. Baselines include EOGS (Savant Aira et al., 2025), SAT-NGP (Billouard et al., 2024), and EO-NeRF (Marí et al., 2023). The first column denote the use of preprocessed data. The two last row reports the performance when training directly with the raw RPCs provided by the satellite vendor and raw panchromatic data.